

Answer Key — Pressure, Winds, Storms, and Cyclones

Final answers with brief reasons.

Objective

- Q1 **(b) area** — pressure is force per unit area.
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- Q2 **(b) 30 Pa** — $P = 90 \div 3 = 30 \text{ Pa}$.
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- Q3 **(b) height** — it depends on the height/depth, not width.
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- Q4 **(b) high to low pressure** — air moves high → low pressure.
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- Q5 **(a) eye** — the eye is the calm, lowest-pressure centre.
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- Q6 **(b) False** — fast winds have reduced (lower) pressure.
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- Q7 **pascal** — pressure is measured in pascals.
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- Q8 **thunderstorm** — that is a thunderstorm.
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Short answer

- Q9 Pressure = force per unit area; $P = 120 \div 4 = 30 \text{ Pa}$.
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- Q10 Liquid pressure rises with height, so a high tank gives strong tap flow; water pressure is greatest near the dam's bottom, so a broad base withstands it.
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- Q11 Sea breeze: sea → land (day); land breeze: land → sea (night).
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Long / application

- Q12 (i) The sharp edge has a smaller area, so for the same force the pressure is higher and it cuts easily.
(ii) Pressing the sucker pushes out air, lowering the inside pressure; the higher outside atmospheric pressure holds it to the wall.
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- Q13 Warm moist air rises from warm ocean water; condensation releases heat, warming the air so it rises more, forming a very low-pressure centre. Surrounding air rushes in and Earth's rotation makes it spin, forming the cyclone. Over land, the warm moist air supply is cut off, so it weakens.
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